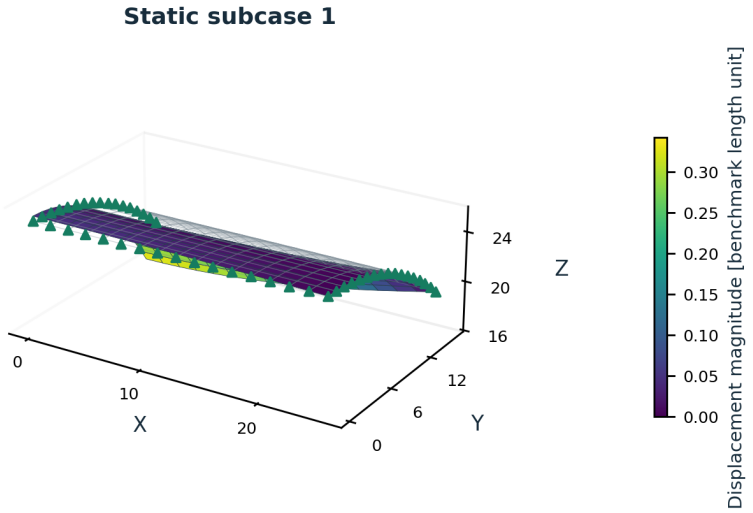
Scordelis-Lo roof

MH_scordelis_lo | SOL 101 | Published shell benchmark

Cylindrical roof quarter model; radius 25, full length 50, thickness 0.25; E=4.32e8, nu=0.0.

Geometry / mesh	25 nodes; 16 elements. Cylindrical roof quarter model; radius 25, full length 50, thickness 0.25; E=4.32e8, nu=0.0.
Material / section	E=4.32e+08, nu=0; density=1; shell thickness=0.25. Consistent benchmark units; displacements in deck length unit, forces in deck force unit, stress in force/length^2
Boundary conditions	Rigid end diaphragm and quarter-symmetry constraints; uniform vertical load 90 per unit surface area.
Applied load	Uniform downward load 90 per unit shell area: density=1, thickness=0.25 and gravity=360 in -Z.



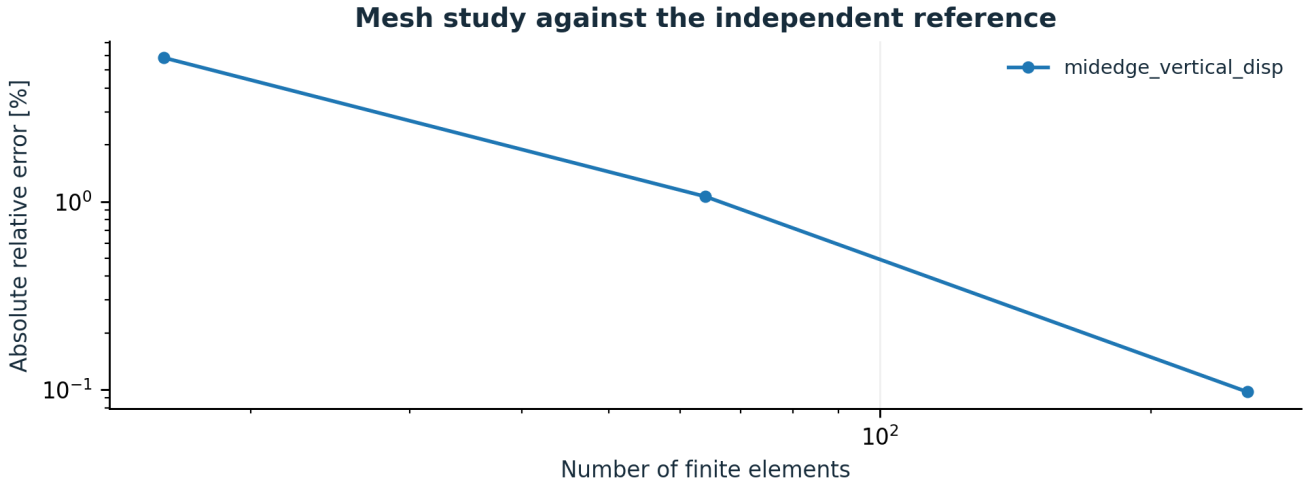
Shape amplified x8.77; grey = original.

Recorded JFEM result; deformation is amplified, mode amplitudes/signs arbitrary. Green triangles mark constrained nodes; magenta arrows mark available point loads. Finest companion shown.

Recorded numerical checks

Quantity [unit]	JFEM	Reference	Error	Assessment
midedge vertical disp [benchmark length unit]	-0.32	-0.3024	5.821%	PASS
N08 refined response [benchmark length unit]	0.30561	0.3024	1.0624%	PASS
N16 refined response [benchmark length unit]	0.30269	0.3024	0.097123%	PASS

Evidence, convergence and interpretation



The figure shows the finest qualified companion; the table retains the supplied benchmark and available refinements. Each field dataset and binary file has its own input hash and run receipt.

Scope: Linear elasticity and small-displacement equilibrium or linearized bifurcation/modal analysis; this is computational verification, not a material-test validation.

Reference and reproduction

MacNeal, R.H.; Harder, R.L. (1985). A proposed standard set of problems to test finite element accuracy. Finite Elements in Analysis and Design 1, 3-20. https://doi.org/10.1016/0168-874X(85)90003-4

Ko, Y.; Lee, Y.; Lee, P.-S.; Bathe, K.-J. (2017). Performance of the MITC3+ and MITC4+ shell elements in widely-used benchmark problems. Computers & Structures 193, 187-206. https://doi.org/10.1016/j.compstruc.2017.08.003

Open the recorded .jfem file with Boxer: Project > Open JFEM results. For older eigenvalue files choose modal or buckling results when requested. The input deck and the recorded run receipt accompany the figures.

Run: 2026-10-19 | Solver revision: a70f997 | Full precision values: results.json / metrics.csv